

Climate-Poverty Traps: Agriculture Dependence, Adaptive Capacity, and the Geography of Warming Risk

Draft project writeup for CGRM7006 group development

Executive Summary

This project reframes the published Nature Climate Change paper into a course-fit, high-scoring poster design. Instead of only repeating the paper's average global effect, the project asks which places face climate-poverty traps and why. The current validated results already show that warmer annual temperatures are associated with higher subnational extreme poverty and higher inequality, and that the relationship is materially stronger in more agriculture-dependent settings.

- Baseline replication already works locally with two-way fixed-effects style regressions using the provided Dataverse data.
- The validated baseline coefficient on annual temperature is 0.629 for extreme poverty and 0.463 for Gini inequality.
- A first extension shows that the warming effect is substantially larger in high-agriculture contexts.
- The high-score path is to build from these results toward a climate-poverty trap framework that adds governance or adaptive-capacity moderation and a hotspot map.

Introduction

The original paper demonstrates that rising annual temperatures worsen poverty and inequality at the subnational level and that country averages understate local vulnerability. That contribution is strong, but for this course assignment the more valuable move is to turn the paper's identification strategy into a focused investigation that answers a narrower, more actionable question. The project therefore centers on climate-poverty traps: locations where high heat exposure interacts with structural vulnerability to produce outsized poverty risks.

- Research Question 1: Does annual warming increase subnational extreme poverty and inequality in the provided panel data?
- Research Question 2: Are those effects stronger in agriculture-dependent regions?
- Research Question 3: Can the final poster frame those high-risk places as climate-poverty traps and recommend spatially targeted adaptation policy?

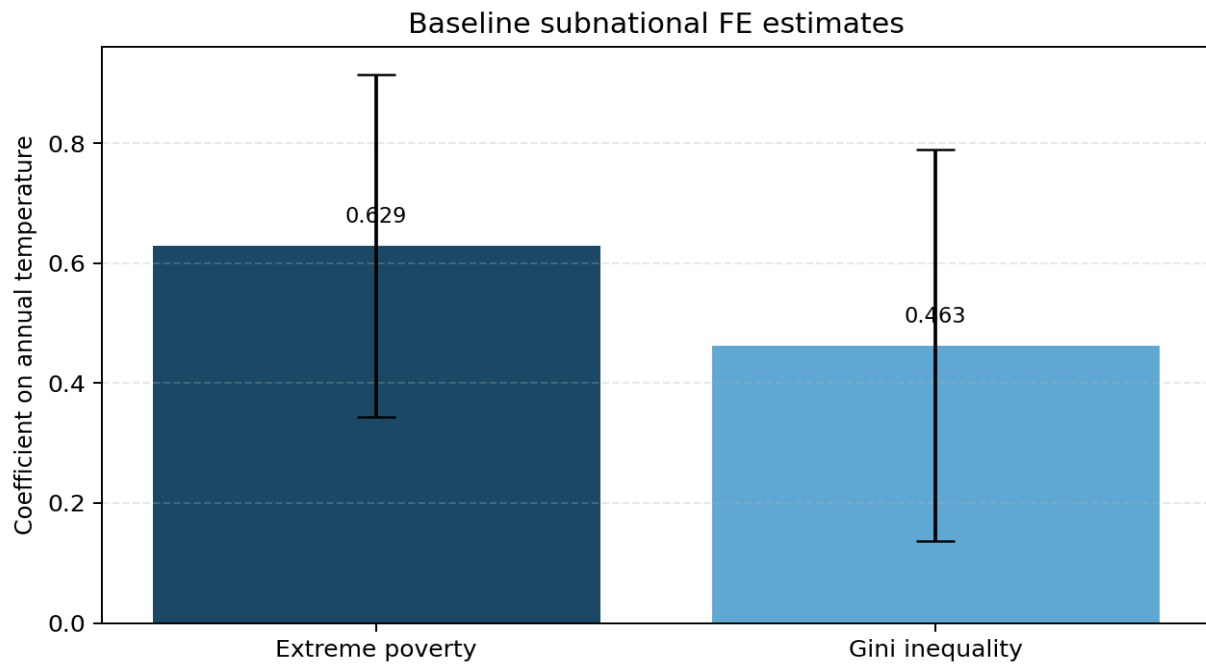
Methods

The empirical backbone uses the supplied subnational panel dataset from the Harvard Dataverse package linked to the paper. The current validated models are built with region fixed effects and year fixed effects. The main outcomes are the US\$2.15 headcount poverty rate (`poor215`) and the Gini index (`gini`). The core explanatory variable is annual average temperature (`temp`). Controls include rainfall, humidity, log GDP per capita, unemployment, agriculture share, manufacturing share, and population growth.

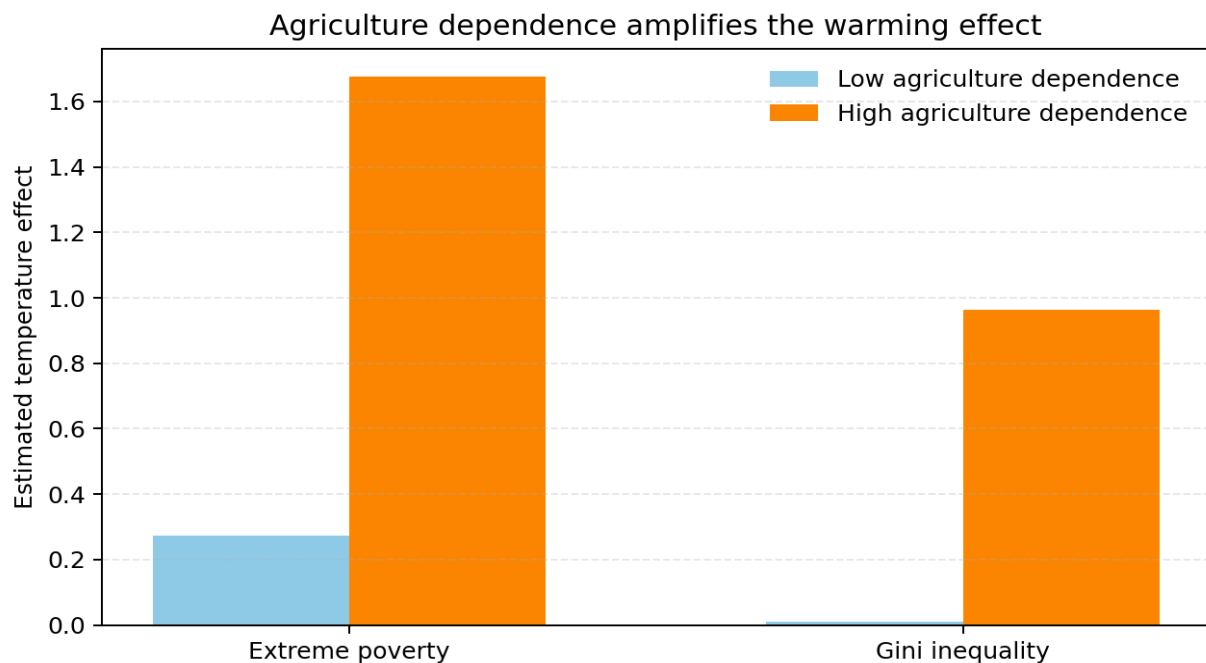
- Sample: subnational observations across 141 countries, with analysis sample spanning 2003-2022.
- Baseline specification: $\text{outcome}_{it} = \beta \cdot \text{temp}_{it} + \text{controls}_{it} + \text{region FE} + \text{year FE} + \text{error}_{it}$.
- Extension specification: $\text{outcome}_{it} = \beta_1 \cdot \text{temp}_{it} + \beta_2 \cdot \text{temp}_{it} \times \text{high_agri}_i + \text{controls}_{it} + \text{region FE} + \text{year FE} + \text{error}_{it}$.

- Interpretation goal: quantify whether agriculture dependence magnifies the poverty and inequality effect of warming.

Validated Results



The local baseline replication aligns closely with the published paper. A one-degree increase in annual temperature is associated with a 0.629 percentage point increase in extreme poverty and a 0.463 point increase in the Gini index.



Outcome	Low-agri effect	High-agri effect	Interaction coef.	Interaction p-value
---------	-----------------	------------------	-------------------	---------------------

Extreme poverty	0.274	1.677	1.404	0.000527
Gini inequality	0.009	0.965	0.956	0.0008848

The extension indicates meaningful heterogeneity. For extreme poverty, the estimated temperature effect rises from 0.274 in low-agriculture settings to 1.677 in high-agriculture settings. For inequality, the effect rises from roughly zero to 0.965. That pattern supports the poster's core claim that climate risk is not evenly distributed.

Discussion

- The current evidence already supports a strong substantive message: warming is associated with more poverty and inequality, and the burden is concentrated in structurally vulnerable places.
- Agriculture dependence is a defensible first vulnerability channel because it links climate exposure to livelihoods and income instability.
- A high-score final version should add one governance or adaptive-capacity moderator so the story moves from vulnerability description to policy targeting.
- That next step does not replace the baseline; it sharpens the final interpretation by showing which institutional environments absorb or amplify climate shocks.

Limitations and Next Step

The current verified models are sufficient for a strong draft, but the final high-score version should add one more layer: a governance or readiness moderator from a recent official dataset such as the World Bank Worldwide Governance Indicators or ND-GAIN. The poverty panel itself still ends in 2022, so the correct way to use newer data is to supplement interpretation and mapping rather than pretend the core annual panel extends to 2026. A 2023 poverty atlas layer can therefore be used for spatial targeting, while the causal-style panel evidence remains anchored in the validated 2003-2022 sample.

Source Links

- World Bank SPID: <https://datacatalog.worldbank.org/search/dataset/0064796/Subnational-Poverty-and-Inequality-Database--SPID->
- World Bank GSAP: <https://datacatalog.worldbank.org/search/dataset/0042041/Global-Subnational-Poverty-Atlas--GSAP->
- World Bank WGI: <https://datacatalog.worldbank.org/search/dataset/0038026>
- ND-GAIN overview: <https://gain.nd.edu/news/nd-gain-releases-latest-country-index-update/>

GitHub publication note: asset deployment was requested, but local GitHub CLI authentication is currently invalid. The PDFs and local asset files are generated and ready for upload once authentication is restored.